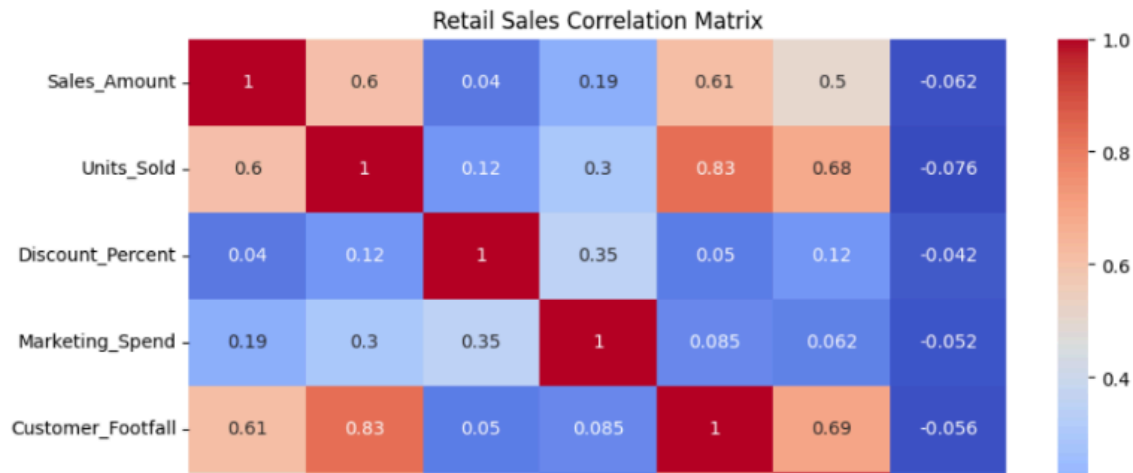




This **Correlation Heatmap** is one of the most important graphs in your VARMAX project. It shows **how strongly one variable is related to another**.

Correlation Value Interpretation

Correlation Value	Relationship
+1.0	Perfect Positive
+0.7 to +0.9	Strong Positive
+0.4 to +0.7	Moderate Positive
+0.1 to +0.4	Weak Positive
0	No Relationship
-0.1 to -0.4	Weak Negative
-0.4 to -0.7	Moderate Negative
-0.7 to -1.0	Strong Negative

Key Findings from Your Heatmap

1. Sales_Amount vs Customer_Footfall

Correlation	Value
-------------	-------

Sales ↔ Footfall	0.61
------------------	------

Meaning

When more customers visit the store, sales increase.

Business Interpretation

✓ More footfall → More sales revenue

2. Units_Sold vs Customer_Footfall

Correlation	Value
-------------	-------

Units Sold ↔ Footfall	0.83
-----------------------	------

Meaning

Very strong positive relationship.

Business Interpretation

✓ Customer footfall is the biggest driver of product sales.

3. Units_Sold vs Online_Orders

Correlation	Value
-------------	-------

Units Sold ↔ Online Orders	0.68
----------------------------	------

Meaning

Online orders significantly contribute to total units sold.

4. Sales_Amount vs Online_Orders

Correlation	Value
Sales ↔ Online Orders	0.50

Meaning

Online sales have a moderate positive impact on revenue.

5. Discount_Percent vs Sales_Amount

Correlation	Value
Discount ↔ Sales	0.04

Meaning

Very weak relationship.

Business Interpretation

Discounts alone are not significantly increasing sales.

6. Marketing_Spend vs Sales_Amount

Correlation	Value
Marketing ↔ Sales	0.19

Meaning

Weak positive relationship.

Business Interpretation

Marketing has some impact, but not as much as customer footfall.

7. Economic_Index

Correlation	Value
Economic Index ↔ Sales	-0.06

Meaning

Almost no relationship.

Business Interpretation

Economic conditions had minimal influence on sales in this dataset.

Most Important Variables

Rank	Variable	Correlation
1	Customer_Footfall ↔ Units_Sold	0.83
2	Customer_Footfall ↔ Online_Orders	0.69
3	Units_Sold ↔ Online_Orders	0.68
4	Sales_Amount ↔ Customer_Footfall	0.61
5	Sales_Amount ↔ Units_Sold	0.60

Viva Answer

Mam: What do you understand from the correlation heatmap?

Answer:

The correlation heatmap shows the relationships among retail sales variables. Customer_Footfall has the strongest positive correlation with Units_Sold (0.83), indicating that increased store visits lead to higher product sales. Sales_Amount is also positively correlated with Customer_Footfall (0.61) and Online_Orders (0.50). Discount_Percent and Economic_Index exhibit weak correlations with sales, suggesting a limited impact on overall revenue. Therefore, customer traffic and online orders are the primary drivers of retail sales in this dataset.

Report Conclusion Table

Variable	Impact on Sales
Customer_Footfall	★★★★★ Very High
Online_Orders	★★★★★ High
Units_Sold	★★★★★ High
Marketing_Spend	★★★ Low
Discount_Percent	★ Very Low
Economic_Index	★ Negligible

Final Business Insight

Customer Footfall and Online Orders are the most influential factors affecting retail sales performance. These variables justify the use of VARMAX because they act as important external predictors for forecasting future sales.

colour identification?

Heatmap Color Identification

Your heatmap uses the **coolwarm** color scheme.

```
sns.heatmap(  
    daily_data.corr(),  
    annot=True,  
    cmap='coolwarm'  
)
```

Color Meaning Table

Color	Correlation Range	Meaning
 Dark Red	+0.80 to +1.00	Very Strong Positive Relationship
 Light Red / Orange	+0.40 to +0.79	Moderate Positive Relationship
 White / Light Grey	Around 0	No Significant Relationship
 Light Blue	-0.10 to -0.40	Weak Negative Relationship
 Dark Blue	-0.40 to -1.00	Strong Negative Relationship

Your Heatmap Examples

Variables	Correlation	Color	Interpretation
Units_Sold ↔ Customer_Footfall	0.83	 Dark Red	Very Strong Positive
Customer_Footfall ↔ Online_Orders	0.69	 Orange/Red	Strong Positive
Sales_Amount ↔ Customer_Footfall	0.61	 Orange	Moderate Positive
Sales_Amount ↔ Discount_Percent	0.04	 Near White	Very Weak Relationship
Economic_Index ↔ Sales_Amount	-0.06	 Light Blue	Very Weak Negative

Viva Answer

Mam: What does the red color indicate?

Red color indicates a positive correlation. The darker the red, the stronger the positive relationship between the two variables.

Mam: What does the blue color indicate?

Blue color indicates a negative correlation. The darker the blue, the stronger the negative relationship between the variables.

Mam: What does white color indicate?

White or very light color indicates little or no relationship between the variables.

Easy Memory Trick

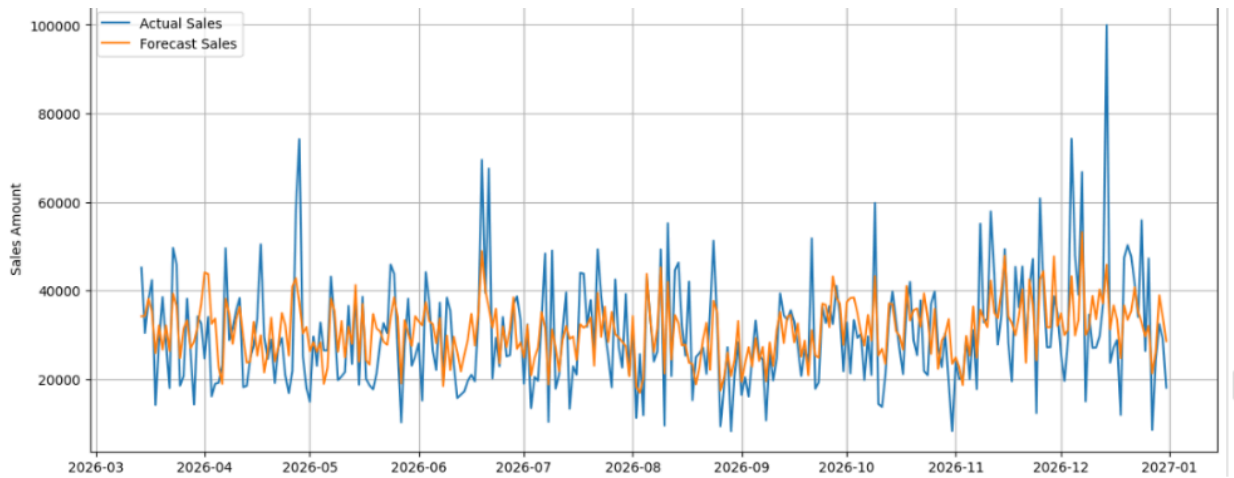
Color	Meaning
 Red	Move Together $\uparrow\uparrow$
 White	No Relationship
 Blue	Move Opposite $\uparrow\downarrow$

Example from Your Project

-  **Units_Sold** $\leftrightarrow$ **Customer_Footfall (0.83)** $\rightarrow$ More customers $\rightarrow$ More products sold.
-  **Sales_Amount** $\leftrightarrow$ **Customer_Footfall (0.61)** $\rightarrow$ More customers $\rightarrow$ More revenue.
-  **Economic_Index** $\leftrightarrow$ **Sales_Amount (-0.06)** $\rightarrow$ Almost no impact on sales.

This is the explanation most faculty members expect when they ask about the colors in a correlation heatmap.

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This graph is called **Actual vs Forecast Sales Graph**.

Graph Components

Line	Color	Meaning
Actual Sales	● Blue	Real sales from the dataset
Forecast Sales	● Orange	Sales predicted by the VARMAX model

What Do You Understand?

1. Forecast Follows the Overall Trend

The orange line generally follows the direction of the blue line.

Meaning:

The VARMAX model has learned the sales pattern and is able to track the overall trend.

2. Forecast is Smoother than Actual Data

Observe:

- Blue line has many sharp spikes.
- Orange line is smoother.

Meaning:

The model captures the average sales behavior but cannot perfectly predict sudden sales jumps.

Example:

Actual Sales	Forecast Sales
100,000	45,000
15,000	25,000

The model smooths extreme values.

3. High Sales Peaks Are Not Fully Captured

Examples around:

May 2026
June 2026
December 2026

Blue spikes reach:

60,000 – 100,000

Orange remains around:

30,000 – 50,000

Meaning:

The model underestimates very high sales events.

Possible reasons:

- Festivals
 - Seasonal demand
 - Flash sales
 - Promotions not fully represented
-

4. No Random Forecasting

If the orange line were completely different from blue:

✗ Bad Model

Since orange roughly follows blue:

✓ Acceptable Model

Project Interpretation Table

Observation	Interpretation
Forecast follows trend	Model learned sales behavior
Forecast smoother	Model captures average pattern
Large spikes missed	Extreme events difficult to predict
Forecast close to actual most of the time	Model performance is reasonable

Viva Answer

Mam: What do you understand from the Actual vs Forecast graph?

Answer:

The Actual vs Forecast graph compares real sales values with the sales predicted by the VARMAX model. The forecast line follows the general trend of the actual

sales data, indicating that the model successfully captures underlying sales patterns. Although the model does not perfectly predict sudden spikes in sales, it provides a reasonable approximation of overall sales behavior and demonstrates the effectiveness of incorporating external factors such as discounts, marketing spend, customer footfall, online orders, and economic conditions.

One-Line Conclusion for Report

The forecasted sales closely follow the overall trend of actual sales, demonstrating that the VARMAX model effectively captures retail sales dynamics while smoothing short-term fluctuations and extreme spikes.

Based on your graph + metrics:

Metric	Result
MAE	7424.84
RMSE	9718.66
Trend Capture	 Good
Extreme Spike Prediction	 Moderate
Overall Model Performance	 Satisfactory

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